Shraddha Takmoge

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PROFESSIONAL SUMMARY

AI/ML Engineer focused on building LLM applications that are measurable, not just demoable. Experienced in RAG pipelines, hybrid retrieval, and agentic workflows using LangChain, LangGraph, and MCP, validated with metrics like Recall@K, MRR, and Faithfulness. Driven to ship reliable, well-tested AI systems.

TECHNICAL SKILLS

- **Programming:** Python, SQL
- **Machine Learning:** Scikit-learn, Regression, Classification, Feature Engineering, Model Evaluation
- **NLP & Deep Learning:** NLP, Neural Networks, Transformers, Embeddings
- **Generative AI:** LLMs, Prompt Engineering, RAG, Self-RAG, CRAG, LangChain, LangGraph, MCP, LoRA Fine-Tuning
- **RAG & Retrieval:** FAISS, Pinecone, Semantic Search, Hybrid Search, BM25, Reciprocal Rank Fusion (RRF)
- **RAG Evaluation:** DeepEval, LLM-as-a-Judge, Precision@K, Recall@K, Hit@K, MRR, NDCG, Context Precision and Recall, Faithfulness, Relevancy
- **Backend & Deployment:** FastAPI, REST APIs, Pydantic, Streamlit, Docker, AWS (EC2, S3), Render
- **Libraries & Tools:** Pandas, NumPy, PyTorch, Hugging Face, LangSmith, Git, GitHub, Jupyter Notebook, Google Colab, VS Code

PROJECTS

RepoLens AI – GitHub Repository Investigator *LangGraph — GitHub MCP — RAG — Pinecone — FastAPI*

- Built a LangGraph-based repository investigation agent using GitHub MCP tools to inspect repository structure, source files, commit history, and answer repository-specific questions through natural language.
- Implemented hybrid code retrieval using Pinecone dense search and BM25, combining semantic and lexical results with Reciprocal Rank Fusion (RRF).
- Developed an agentic workflow using LangGraph and FastMCP to interact with GitHub through repository analysis, file retrieval, and code search tools.
- Evaluated retrieval on 20 repository-specific queries using Hit@K and Recall@K, achieving 55% Hit@1, 75% Hit@3, 80% Hit@4, and 68.33% Recall@4; removed reranking after evaluation showed weaker retrieval performance.

AI Decision Auditor *Python — LangChain — Pinecone — Groq — Ollama — DeepEval — Pydantic*

- Built a RAG-based AI decision auditing system with LangChain, Pinecone retrieval, and GPT-OSS 120B for evidence-grounded answer generation.
- Implemented and evaluated document retrieval using Precision@K, Recall@K, NDCG@K, and MRR, achieving 1.00 Precision@1, 1.00 Recall@3, and 1.00 MRR across 10 queries.
- Evaluated contextual retrieval with DeepEval and Gemma 3 4B, achieving 0.952 Contextual Precision and 1.00 Contextual Recall across 5 queries.
- Evaluated the generator using Faithfulness, Answer Relevancy, and G-Eval Correctness, with an end-to-end RAG pipeline evaluation using the actual retriever and generator.

EXPERIENCE

Data Science Intern – Rajavi Technology Pvt. Ltd. *Dec 2024 – Jan 2025*

- Built a content-based movie recommendation system in Python using the TMDB 5000 dataset (~4,800 movies), merging and cleaning movie and credits data with Pandas.
- Vectorized movie tags with CountVectorizer (5000 features, stopword removal) and NLTK's PorterStemmer, then used cosine similarity to recommend the top 5 most similar movies for a given title; saved the model using Pickle.

Data Science Intern – YHills *Nov 2023 – Dec 2023*

- Applied Pandas and NumPy for data manipulation and preprocessing, including cleaning and transforming raw datasets for analysis.
- Practiced feature engineering and data preprocessing techniques to prepare data for machine learning models.

EDUCATION

Savitribai Phule Pune University *Nov 2022 – Jun 2026*

Bachelor of Engineering – Artificial Intelligence and Data Science
Overall CGPA: 9.25 / 10.00

ACHIEVEMENTS

- Selected for the **Women in Engineering (WIE) Program** conducted by **Shikshana Foundation** in collaboration with the **AMADEUS** team.
- **Runner-up** at the **ISDM Code4Change Hackathon**